Advanced Power System Analysis

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1

2 Introduction

In this document, I review two papers. The first papers introduces the derivation is for three-phase, balanced systems. The second derviation is for three-phase unbalanced systems.

2.1 First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al.

2.2 Paper's Objectives

1- How can we change the network configuration to reduce resistive losses? 2- How can the load be distributed such that no overloads occur on the T-Lines?

3 Background Information

NOTE: This file contains supporting matrix algebra and the fuller multiphase branch-flow formulation of Gan and Low (Gan and Low 2014). The working equations are in `ch2_voltage_reactive_equations.qmd`.

3.1 Three-phase Branch Power Matrix

Recall that $\bar{\mathbf{V}}_i$ and $\bar{\mathbf{I}}_{ij}$ are 3×1 column vectors. We previously used the per-phase complex power vector:

$$\bar{\mathbf{s}}_{ij} = \begin{bmatrix} P_{ij}^a + jQ_{ij}^a \\ P_{ij}^b + jQ_{ij}^b \\ P_{ij}^c + jQ_{ij}^c \end{bmatrix}$$

Gan and Low also use the full branch power matrix:

$$\bar{\mathbf{S}}_{ij} = \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \tag{3.1}$$

Expanding the matrix multiplication:

$$\bar{\mathbf{S}}_{ij} = \begin{bmatrix} \bar{V}_i^a (\bar{I}_{ij}^a)^* & \bar{V}_i^a (\bar{I}_{ij}^b)^* & \bar{V}_i^a (\bar{I}_{ij}^c)^* \\ \bar{V}_i^b (\bar{I}_{ij}^a)^* & \bar{V}_i^b (\bar{I}_{ij}^b)^* & \bar{V}_i^b (\bar{I}_{ij}^c)^* \\ \bar{V}_i^c (\bar{I}_{ij}^a)^* & \bar{V}_i^c (\bar{I}_{ij}^b)^* & \bar{V}_i^c (\bar{I}_{ij}^c)^* \end{bmatrix}$$

Note that the **diagonal elements** are the conventional complex powers for each phase:

$$\text{diag}(\bar{\mathbf{S}}_{ij}) = \bar{\mathbf{s}}_{ij}$$

The **off-diagonal elements** are cross-phase voltage-current products. They do not represent extra downstream branches or separate real/reactive phase powers. The full 3×3 matrix is useful for Gan and Low's branch-flow and optimization formulations, but it is not required just to substitute phase power into KVL.

3.2 Derive the Magnitude of the Voltage

Following the same approach as in Equation 4.21, we can obtain squared voltage magnitudes using the three-phase KVL equation:

$$\bar{\mathbf{V}}_j = \bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij}$$

To retain all cross-phase voltage products, multiply the equation on the right by its **Hermitian transpose**:

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})^H \tag{3.2}$$

The Hermitian transpose, denoted by H, means **conjugate transpose**; it does **not** mean extracting the diagonal. A 3×1 voltage vector times its 1×3 conjugate transpose produces a 3×3 matrix.

3.2.1 Solving for the left hand side of Equation 3.2

$$\bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H = \begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix}_{3\times1} \begin{bmatrix} (\bar{V}_j^a)^* & (\bar{V}_j^b)^* & (\bar{V}_j^c)^* \end{bmatrix}_{1\times3} = \begin{bmatrix} |\bar{V}_j^a|^2 & \bar{V}_j^a(\bar{V}_j^b)^* & \bar{V}_j^a(\bar{V}_j^c)^* \\ \bar{V}_j^b(\bar{V}_j^a)^* & |\bar{V}_j^b|^2 & \bar{V}_j^b(\bar{V}_j^c)^* \\ \bar{V}_j^c(\bar{V}_j^a)^* & \bar{V}_j^c(\bar{V}_j^b)^* & |\bar{V}_j^c|^2 \end{bmatrix}_{3\times3} \tag{3.3}$$

Note that the diagonal elements are **squared** voltage magnitudes. The off-diagonal elements remain complex cross-phase voltage products.

Gan and Low define the voltage matrix as:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_j\bar{\mathbf{V}}_j^H \tag{3.4}$$

Similarly:

$$\bar{\mathbf{W}}_i = \bar{\mathbf{V}}_i\bar{\mathbf{V}}_i^H \tag{3.5}$$

3.2.2 Solving for the right hand side of Equation 3.2

We can apply the same Hermitian transpose to the right-hand side:

$$\bar{\mathbf{W}}_j = (\bar{\mathbf{V}}_i - \bar{\mathbf{Z}}_{ij}\bar{\mathbf{I}}_{ij})(\bar{\mathbf{V}}_i^H - \bar{\mathbf{I}}_{ij}^H\bar{\mathbf{Z}}_{ij}^H)$$

Voltage Eq. Note

$$(\bar{\mathbf{Z}}\bar{\mathbf{I}})^H = \bar{\mathbf{I}}^H \bar{\mathbf{Z}}^H$$

The multiplication order reverses because $(AB)^H = B^H A^H$. This is a property of the conjugate transpose, not merely a rule for fixing dimensions.

Expanding the right-hand side:

$$\bar{\mathbf{W}}_j = \bar{\mathbf{V}}_i \bar{\mathbf{V}}_i^H - \bar{\mathbf{V}}_i \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{V}}_i^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H \bar{\mathbf{Z}}_{ij}^H \tag{3.6}$$

We previously defined the voltage matrix in Equation 3.5 and the branch-power matrix in Equation 3.1. Accordingly, we can define the current outer-product matrix as:

$$\bar{\mathbf{L}}_{ij} = \bar{\mathbf{I}}_{ij} \bar{\mathbf{I}}_{ij}^H$$

Therefore, Equation 3.6 can be written as:

$$\boxed{\bar{\mathbf{W}}_j = \bar{\mathbf{W}}_i - \bar{\mathbf{S}}_{ij} \bar{\mathbf{Z}}_{ij}^H - \bar{\mathbf{Z}}_{ij} \bar{\mathbf{S}}_{ij}^H + \bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H} \tag{3.7}$$

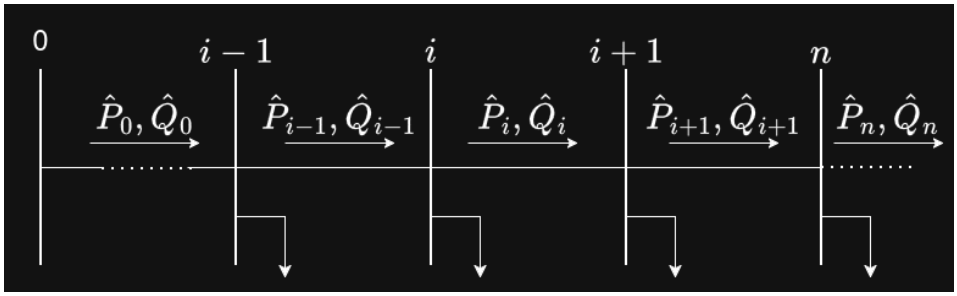
Note

Equation 3.7 is an exact squared-voltage **matrix** relationship derived from KVL. The term $\bar{\mathbf{Z}}_{ij} \bar{\mathbf{L}}_{ij} \bar{\mathbf{Z}}_{ij}^H$ is quadratic in line current; it is the outer product of the line voltage-drop vector with itself. Its diagonal is not the same as the real-power loss. For our immediate voltage–reactive-power work, the per-phase power equations and KVL are sufficient as starting equations, so we can keep this fuller derivation for background.

4 Equations Derivation

NOTE: Baran and Wu use the DistFlow equations for a balanced, single-phase equivalent radial distribution system. We use these equations as a starting point for our unbalanced three-phase work.

Consider the following single line diagram:



(a) ht

Figure 4.1: A single Line Diagram (SLD) for an n bus system. The loads are unknown for now. The complex power ($\bar{S}$) flowing between each bus is shown.

For each load, we consider $P_{L,n}$ and $Q_{L,n}$, indicating real and reactive power consumed at bus n. Positive load means consumption.

Each line has an impedance, $\bar{Z}_n = r_n + jx_n$. The sending-end complex power flow on each line is given by:

$$\bar{S_n} = P_n + jQ_n$$

For instance, for the line leaving bus $i+1$:

$$\bar{S}_{i+1} = P_{i+1} + jQ_{i+1}$$

4.1 Deriving the Real and Reactive Power Equations

I think it is easier to see this if we look at the power flow between two consecutive buses. Consider the power flow between buses n and $n+1$. We define P_n as the power leaving bus n and entering line $(n, n+1)$:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{P_{n,loss}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.1}$$

4.1.1 Decomposing the Losses term in equation Equation 4.1

$$P_{loss} = r|I|^2$$

At any given bus n:

$$P_{n,loss} = r_n|I_n|^2 \tag{4.2}$$

Recall:

$$\bar{S} = \bar{V}\bar{I}^* = P + jQ$$

Wherein the magnitude of the apparent power is given by:

$$|S| = |I|\ |V| = \sqrt{P^2 + Q^2}$$

Squaring both sides:

$$|S|^2 = |I|^2\ |V|^2 = P^2 + Q^2$$

Solving for the squared current magnitude:

$$|I|^2 = \frac{P^2 + Q^2}{|V|^2}$$

For the line leaving bus n, the squared current magnitude is:

$$|I_n|^2 = \frac{P_n^2 + Q_n^2}{|V_n|^2} \tag{4.3}$$

substituting the current equation Equation 4.3 into Equation 4.2 we get:

$$P_{n,loss} = r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2} \tag{4.4}$$

Substituting Equation 4.4 in the power balance between buses n and $n+1$ in Equation 4.1, we get:

$$\underbrace{P_n}_{\text{Power leaving bus } n} = \underbrace{r_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Line loss}} + \underbrace{P_{n+1}}_{\text{Power leaving bus } n+1} + \underbrace{P_{L,n+1}}_{\text{Load at bus } n+1} \tag{4.5}$$

> **Note**
>
> Equation 4.5 is the real power DistFlow Equation. This is the exact branch real-power balance for the single-phase equivalent line model, assuming the bus has one downstream line and no additional injection. For a branching feeder, we also account for every outgoing branch.

Similarly, the reactive power can be calculated in a similar manner:

$$\underbrace{Q_n}_{\text{Q leaving bus } n} = \underbrace{x_n \frac{P_n^2 + Q_n^2}{|\bar{V}_n|^2}}_{\text{Reactive power absorbed in the line}} + \underbrace{Q_{n+1}}_{\text{Q leaving bus } n+1} + \underbrace{Q_{L,n+1}}_{\text{Q consumed by bus } n+1 \text{ load}} \tag{4.6}$$

4.2 Deriving the Voltage Equation:

Applying KVL to buses, i and $i-1$ in Figure 4.1:

$$\bar{V}_{i-1} = \bar{Z}_{i-1}\ \bar{I}_{i-1} + \bar{V}_i \tag{4.7}$$

The current complex value can be solved for in a similar manner as in equation Equation 4.3:

$$\bar{V}_n\ \bar{I}_n^* = P_n + jQ_n$$

Taking the complex conjugate and solving for $\bar{I}_n$:

$$\bar{I}_n = \frac{P_n - jQ_n}{\bar{V}_n^*} \tag{4.8}$$

Substituting Equation 4.8 in Equation 4.7 & solving for $\bar{V}_i$:

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1} \tag{4.9}$$

$$\bar{V}_i = \bar{V}_{i-1} - \bar{Z}_{i-1}\ \frac{P_{i-1} - jQ_{i-1}}{\bar{V}^*_{i-1}}$$

We can then extend this to get only the magnitude of the voltage:

Recall $|a-b|^2 = (a-b)(a^*-b^*)$:

$$|\bar{V}_i|^2 = (\bar{V}_{i-1} - \bar{Z}_{i-1}\ \bar{I}_{i-1})\ (\bar{V}^*_{i-1} - \bar{Z}^*_{i-1}\ \bar{I}^*_{i-1})$$

Multiplying both terms, we end up with:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2\ \Re\{\underbrace{\bar{V}^*_{i-1}\ \bar{I}_{i-1}}_{\substack{\text{Apparent}\\ \text{Power}\\ \text{Conjugate}}}\ \bar{Z}_{i-1}\} + |\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 \tag{4.10}$$

Recall $\bar{S}^*_{i-1} = \bar{V}^*_{i-1}\ \bar{I}_{i-1} = P_{i-1} - jQ_{i-1}$

So the middle term can be decomposed such that:

$$\bar{V}^*_{i-1}\ \bar{I}_{i-1}\ \bar{Z}_{i-1} = (P_{i-1} - jQ_{i-1})(r_{i-1} + jx_{i-1})$$

Multiplying and obtaining only the real part because we are trying to calculate the magnitude:

$$\Re\{\bar{Z}_{i-1}\ \bar{V}^*_{i-1}\ \bar{I}_{i-1}\} = r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1} \tag{4.11}$$

The last term in Equation 4.10 can be simplified further. Recall that $|\bar{Z}_{i-1}|^2 = r^2_{i-1} + x^2_{i-1}$ and $|\bar{I}_{i-1}|^2 = \frac{P^2_{i-1}+Q^2_{i-1}}{|\bar{V}_{i-1}|^2}$

$$|\bar{Z}_{i-1}|^2\ |\bar{I}_{i-1}|^2 = (r^2_{i-1} + x^2_{i-1})\ (\frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2}) \tag{4.12}$$

Substituting Equation 4.11 and Equation 4.12 in Equation 4.10:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}) + (r^2_{i-1} + x^2_{i-1})\ \frac{P^2_{i-1} + Q^2_{i-1}}{|\bar{V}_{i-1}|^2} \tag{4.13}$$

Note that Equation 4.13 is essentially:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}) + \underbrace{(r_{i-1}^2 + x_{i-1}^2)}_{\text{Imedance } Z^2} \underbrace{\frac{P_{i-1}^2 + Q_{i-1}^2}{|\bar{V}_{i-1}|^2}}_{\text{Current } I^2}$$

$V = ZI$, making the last term a loss term. This loss term is a second order contribution associated with line losses. Assuming the losses insignificant, we can drop this term and the equation becomes:

$$|\bar{V}_i|^2 = |\bar{V}_{i-1}|^2 - 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})$$

Rearranging the equation:

$$(|\bar{V}_i|^2 - |\bar{V}_{i-1}|^2) \approx 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})$$

Recall $a^2 - b^2 = (a-b)(a+b)$:

$$(|\bar{V}_i| - |\bar{V}_{i-1}|)(|\bar{V}_i| + |\bar{V}_{i-1}|) \approx 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})$$

Defining the voltage drop as: $\Delta V = |\bar{V}_{i-1}| - |\bar{V}_i|$:

$$\Delta V(|\bar{V}_i| + |\bar{V}_{i-1}|) \approx 2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})$$

Solving for ΔV:

$$\Delta V \approx \frac{2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})}{(|\bar{V}_i| + |\bar{V}_{i-1}|)}$$

Assuming the voltage difference between adjascent buses is small:

$$|\bar{V}_i| \approx |\bar{V}_{i-1}| \approx V$$

Therefore,

$$|\bar{V}_i| + |\bar{V}_{i-1}| \approx 2V$$

substituting in the previous ΔV equation:

$$\Delta V \approx \frac{2(r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1})}{2V}$$

Cancelling the 2:

$$\Delta V \approx \frac{r_{i-1}\ P_{i-1} + x_{i-1}\ Q_{i-1}}{V} \tag{4.14}$$

> **Note**
>
> Equation 4.13 is the Voltage Magnitude DistFlow Equation. This equation applies exactly to each branch under the paper's single-phase equivalent series-impedance model. It does not, by itself, describe mutual coupling in an unbalanced three-phase system.

4.3 Three-phase Unbalanced System

Using the same KVL we used before, Equation 4.9, we can set a system of equations for buses i and j such that i is upstream of j:

$$\begin{bmatrix} \bar{V}_j^a \\ \bar{V}_j^b \\ \bar{V}_j^c \end{bmatrix} = \begin{bmatrix} \bar{V}_i^a \\ \bar{V}_i^b \\ \bar{V}_i^c \end{bmatrix} - \begin{bmatrix} \bar{Z}_{ij}^{aa} & \bar{Z}_{ij}^{ab} & \bar{Z}_{ij}^{ac} \\ \bar{Z}_{ij}^{ba} & \bar{Z}_{ij}^{bb} & \bar{Z}_{ij}^{bc} \\ \bar{Z}_{ij}^{ca} & \bar{Z}_{ij}^{cb} & \bar{Z}_{ij}^{cc} \end{bmatrix} \begin{bmatrix} \bar{I}_{ij}^a \\ \bar{I}_{ij}^b \\ \bar{I}_{ij}^c \end{bmatrix} \tag{4.15}$$

Note that the diagonal elements of $\bar{Z}_{ij}$ are self-impedances, and the off-diagonal elements are mutual impedances. This is a three-phase line model; a line with fewer phases uses the appropriate reduced impedance matrix. Neutral and transformer modeling must be handled consistently in the feeder model.

For any given phase, the voltage equation can be extracted from the above matrix form. For instance:

$$\bar{V}_j^a = \bar{V}_i^a - \bar{Z}_{ij}^{aa}\bar{I}_{ij}^a - \bar{Z}_{ij}^{ab}\bar{I}_{ij}^b - \bar{Z}_{ij}^{ac}\bar{I}_{ij}^c \tag{4.16}$$

We can rewrite the current terms in Equation 4.16 using complex power for each phase. At the sending end of line (i, j):

$$\bar{S}_{ij}^a = \bar{V}_i^a(\bar{I}_{ij}^a)^* = P_{ij}^a + jQ_{ij}^a$$
$$\bar{S}_{ij}^b = \bar{V}_i^b(\bar{I}_{ij}^b)^* = P_{ij}^b + jQ_{ij}^b$$

$$\bar{S}_{ij}^{c} = \bar{V}_{i}^{c}(\bar{I}_{ij}^{c})^{*} = P_{ij}^{c} + jQ_{ij}^{c}$$

As a vector format, we can express the three complex power equations as:

$$\begin{bmatrix} \bar{S}_{ij}^{a} \\ \bar{S}_{ij}^{b} \\ \bar{S}_{ij}^{c} \end{bmatrix} = \begin{bmatrix} \bar{V}_{i}^{a}(\bar{I}_{ij}^{a})^{*} \\ \bar{V}_{i}^{b}(\bar{I}_{ij}^{b})^{*} \\ \bar{V}_{i}^{c}(\bar{I}_{ij}^{c})^{*} \end{bmatrix} = \begin{bmatrix} P_{ij}^{a} + jQ_{ij}^{a} \\ P_{ij}^{b} + jQ_{ij}^{b} \\ P_{ij}^{c} + jQ_{ij}^{c} \end{bmatrix} \tag{4.17}$$

> **Note**
>
> Equation 4.17 is a 3×1 vector of **per-phase complex powers**. It is not equal to a 3×3 matrix. The full branch-power matrix used by Gan and Low, including cross-phase products, is discussed in the background file. Mutual coupling is already present in Equation 4.15 through the off-diagonal impedances.

4.3.1 Derive the current equations

Consider the complex power for phase a between two buses:

$$\bar{S}_{ij}^{a} = \bar{V}_{i}^{a}(\bar{I}_{ij}^{a})^{*}$$

Take the complex conjugate of both sides:

$$\left(\bar{S}_{ij}^{a}\right)^{*} = (\bar{V}_{i}^{a})^{*}\bar{I}_{ij}^{a}$$

Solving for $\bar{I}_{ij}^{a}$:

$$\bar{I}_{ij}^{a} = \frac{\left(\bar{S}_{ij}^{a}\right)^{*}}{(\bar{V}_{i}^{a})^{*}}$$

Which can be rewritten as:

$$\bar{I}_{ij}^{a} = \frac{P_{ij}^{a} - jQ_{ij}^{a}}{(\bar{V}_{i}^{a})^{*}} \tag{4.18}$$

Solving for phases b and c using the same approach:

$$\bar{I}_{ij}^{b} = \frac{P_{ij}^{b} - jQ_{ij}^{b}}{(\bar{V}_{i}^{b})^{*}} \tag{4.19}$$

$$\bar{I}_{ij}^{c} = \frac{P_{ij}^{c} - jQ_{ij}^{c}}{(\bar{V}_{i}^{c})^{*}} \tag{4.20}$$

Equation 4.18, Equation 4.19, and Equation 4.20 can be used in our KVL equation Equation 4.15, such that:

$$\begin{bmatrix} \bar{V}_{j}^{a} \\ \bar{V}_{j}^{b} \\ \bar{V}_{j}^{c} \end{bmatrix} = \begin{bmatrix} \bar{V}_{i}^{a} \\ \bar{V}_{i}^{b} \\ \bar{V}_{i}^{c} \end{bmatrix} - \bar{Z}_{ij} \begin{bmatrix} \dfrac{P_{ij}^{a} - jQ_{ij}^{a}}{(\bar{V}_{i}^{a})^{*}} \\ \dfrac{P_{ij}^{b} - jQ_{ij}^{b}}{(\bar{V}_{i}^{b})^{*}} \\ \dfrac{P_{ij}^{c} - jQ_{ij}^{c}}{(\bar{V}_{i}^{c})^{*}} \end{bmatrix} \tag{4.21}$$

Equation 4.21 expresses the receiving-end voltage phasors as a function of the sending-end voltage phasors, branch impedance, and phase real and reactive power flows (Gan and Low 2014):

$$\boxed{\bar{\mathbf{V}}_{j} = \mathbf{f}(\bar{\mathbf{V}}_{i}, \bar{\mathbf{Z}}_{ij}, \mathbf{P}_{ij}, \mathbf{Q}_{ij})}$$

> **Note**
>
> This is an **exact branch relationship**, not yet an explicit voltage-magnitude sensitivity to inverter reactive power injection. To connect an inverter at bus k to voltages throughout the feeder, we must also use network current/power balance and the inverter injection convention. Gan and Low's squared-voltage matrix derivation is in the background file for reference.

5 Motivation

Distribution voltage is conventionally held near nominal along the primary feeder, typically within ±5%, so that every customer receives a service voltage inside ANSI C84.1 Range A (114–126 V). The limit, however, applies at the customer service point, not on the primary. If residential inverters can correct the voltage locally at each household, the primary feeder voltage no longer has to be chosen to protect the worst-placed customer. It becomes a decision variable: it can be lowered for conservation voltage reduction (CVR) or raised to reduce losses, while inverters hold each household within its service limit.

5.1 State of the Art

- **CVR with smart inverters.** Smart inverters have been coordinated with tap changers, regulators and capacitors to achieve CVR on three-phase unbalanced feeders (**jha2019bilevel?**; **zhang2019distributed?**), including integrated primary-secondary networks with voltage-dependent loads (**zhang2021primarysecondary?**). In these studies inverters *support* CVR, while the primary voltage stays within its conventional band.
- **Grid-edge voltage control.** Dedicated utility-owned devices on the secondary side of service transformers regulate customer voltage so that the primary can be lowered further. Field deployments report up to 5% CVR (**moghe2016gridedge?**), limiting-node voltage increases of 0.88–2.04% (**jha2023gridedge?**), and energy savings up to 4.7% when combined with smart inverters (**padullaparti2023cvr?**).
- **Volt-VAR curve design.** Per-inverter Volt-VAR rules have been optimized for single-phase feeders (**murzakhanov2024optimal?**), unbalanced multiphase feeders (**gupta2025regulating?**), and a single utility-scale battery (**ingebrigtsen2026optimizing?**). All aim to keep voltages near nominal rather than to compensate a deliberately shifted primary.
- **Cross-phase coupling.** On unbalanced four-wire networks, reactive power injected on one phase changes the voltages on the other phases (**wang2020interphase?**; **dalal2024crossphase?**).
- **Siting and sizing.** Methods exist to choose how many voltage-support units are needed and where, but for balanced networks (**giannitrapani2017optimal?**) or PV at medium voltage (**chen2022optimal?**).

5.2 Research Gap

Existing work either regulates customer voltage at the grid edge with dedicated utility devices, or uses residential inverters as auxiliary support while the primary voltage remains within its conventional band. No study has treated the primary feeder voltage as a free decision variable whose feasible range is set by the corrective capability of customer-owned residential inverters.

In particular, it is not known:

1. how far the primary voltage can be shifted, up or down, before participating inverters can no longer hold every household within its service-voltage limit;
2. how this range depends on the number, location and phase of participating inverters, and how it is reduced by cross-phase coupling, non-participating customers and battery operating state on unbalanced feeders;
3. whether the energy saved by operating the primary away from nominal outweighs the additional losses caused by the reactive power needed to compensate it; and
4. whether the required compensation can be delivered autonomously through location-dependent Volt-VAR settings.

We define the **decoupling range** as the set of primary-side voltage set-points for which the participating inverters can keep every household service voltage within its target limits.

5.3 Research Goal

To develop and evaluate a methodology for inverter-enabled grid-edge voltage regulation, in which residential inverters compensate their local service voltage so that the primary feeder voltage can be decoupled from customer voltage limits on unbalanced distribution networks.

5.4 Research Objectives

- **O1 — Quantify the decoupling range.** Develop a sensitivity-based model that determines the decoupling range as a function of inverter count, location, phase assignment and rating.
- **O2 — Identify the limiting factors.** Characterize how cross-phase coupling, non-participating customers and battery operating state (charging, discharging, idle) narrow or extend the decoupling range.
- **O3 — Evaluate the net benefit.** Determine the primary voltage set-point that maximizes net benefit across loading conditions, accounting for voltage-dependent load reduction, additional network losses, transformer loading and substation power factor.

- **O4 — Enable decentralized implementation.** Translate the required reactive power into location-dependent Volt-VAR settings (reference voltage, deadband, slope and reactive limits). Identify when autonomous operation is sufficient and when coordination is needed.

5.5 Research Questions

RQ1 — Decoupling feasibility: For a given set of residential inverters, what range of primary voltage set-points can be supported while keeping every household within its target service voltage, and how does this range scale with inverter count, location, phase assignment and rating?

RQ2 — Limiting factors: How do cross-phase coupling, non-participating customers and battery operating state constrain the decoupling range on unbalanced feeders?

RQ3 — Net benefit: Within the feasible range, which primary voltage set-point maximizes net energy benefit, after netting voltage-dependent load reduction against additional losses, and how does this optimum change with loading?

RQ4 — Decentralized implementation: Can the reactive power required at each location be encoded in Volt-VAR curves that hold household voltages at their targets without communication, and under what conditions does coordination become necessary?

5.6 Scope and Assumptions

- Unbalanced three-phase feeders with service transformers and single-phase residential connections. *To decide: test feeder (e.g., IEEE 8500-node or a PNNL taxonomy feeder) and whether secondaries are modelled as split-phase.*
- Customer-owned single-phase residential inverters (PV and battery) with reactive capability limited by IEEE 1547-2018.
- Voltage-dependent load models (ZIP or CVR factor), since CVR savings depend directly on them.
- Quasi-static time-series simulation over representative loading conditions.

5.7 Expected Contributions

1. The concept and quantification of a **decoupling range** for inverter-enabled grid-edge voltage regulation, and its dependence on inverter participation, location and phase.
2. An analysis of how **cross-phase coupling and battery operating state** limit grid-edge compensation on unbalanced feeders.

3. A **net-benefit assessment** identifying when shifting the primary voltage is worthwhile once added losses are included.
4. A method for deriving **location-dependent Volt-VAR settings** that implement the approach autonomously. Volt-VAR curves are the implementation of the methodology, not its primary objective.

5.8 References

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6 Setup

The GridLAB-D file is set to 15 houses, with a single substation transformer and distribution transformer. There are two nodes: upstream node and node 4. The upstream node is a swing bus node. Node 4 is the main node wherein all the loads connect to.

6.0.1 Initial Case

None of the houses have inverters in them. The voltagge behavior is depicted at each house, as well as the source node.

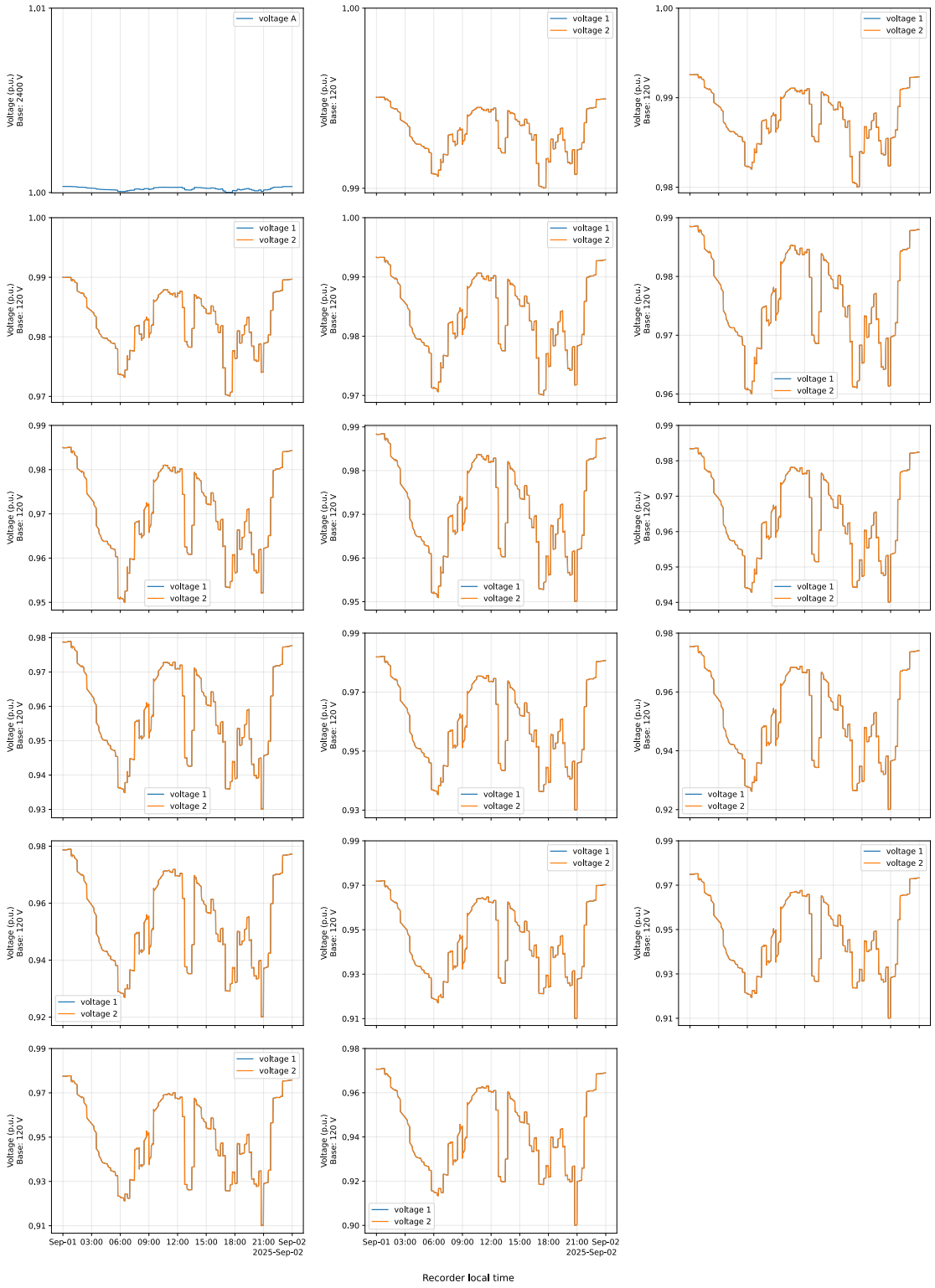


Figure 6.1: The voltage at each house, with a total of 15 houses. The first figure shows the voltage at Node 4, which is not a Swing node (obviously!)

Note that the voltage drops as the house is at a further distance from node 4.

6.0.2 Sweep the voltage at Node 4

When the voltage is swept at Node 4, the same voltage drop is depicted at different houses location.

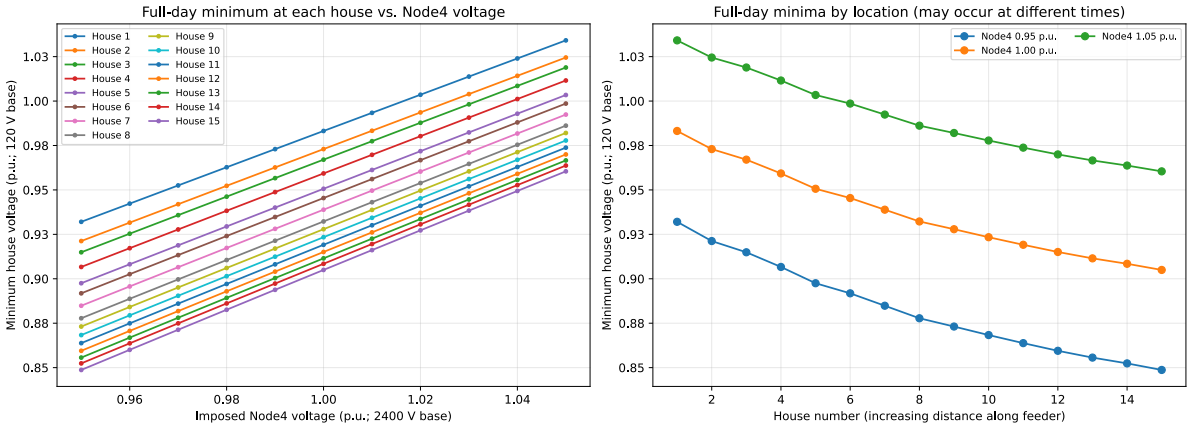


Figure 6.2: Voltage Sweeping Across Node4

Table 6.1: Simulation Results Data

node4_pu	house	min_pu	max_pu	min_timestamp	samples
0.95	10	0.868333	0.927508	2025-09-01 20:46:00 PDT	1439
0.95	11	0.863792	0.926525	2025-09-01 20:46:00 PDT	1439
0.95	12	0.859475	0.92555	2025-09-01 20:46:00 PDT	1439
0.95	13	0.85565	0.925008	2025-09-01 20:46:00 PDT	1439
0.95	14	0.852392	0.924492	2025-09-01 20:46:00 PDT	1439
0.95	15	0.8487	0.924225	2025-09-01 20:46:00 PDT	1439
0.95	1	0.932058	0.94515	2025-09-01 17:30:00 PDT	1439
0.95	2	0.921233	0.942225	2025-09-01 17:28:00 PDT	1439
0.95	3	0.914917	0.939467	2025-09-01 17:28:00 PDT	1439
0.95	4	0.906658	0.936867	2025-09-01 06:29:00 PDT	1439

Figure 6.2 demonstrates that the household voltage response depends on both the upstream voltage and the household's location along the feeder. Note that the distance of voltage lines in Figure 6.2 decreases consecutively. The current flow in the line supplying house 1 carries the combined current of many downstream loads. The line between house 1 and house 2 carries slightly less current, and so forth. From ohm's law:

$$\Delta\bar{\mathbf{V}} = \bar{\mathbf{Z}}\bar{\mathbf{I}}$$

If the line segments approximately equal impedances, the voltage drop across each segment generally decreases as we move downstream because the current decreases. Therefore:

$$|\Delta\bar{\mathbf{V}}_{1,2}| > |\Delta\bar{\mathbf{V}}_{13,14}| > |\Delta\bar{\mathbf{V}}_{14,15}|$$

To explain this further, let's look at all households at their lowest point, which is at 8:46 p.m.

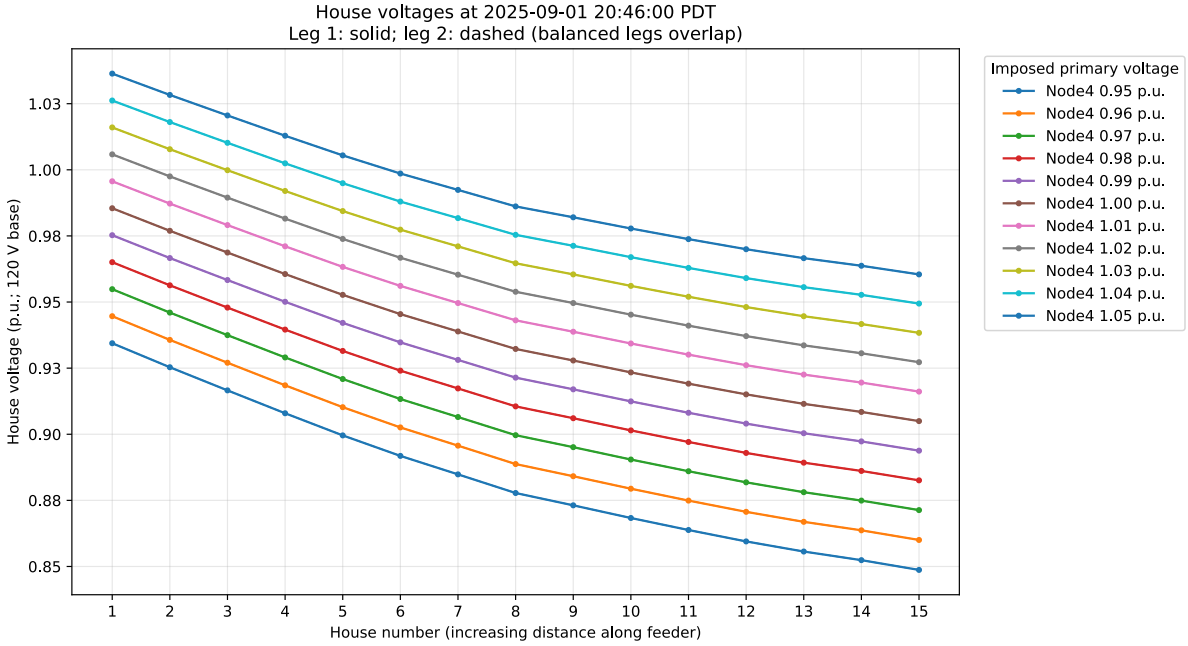


Figure 6.3: Voltage Sweep at 8:46 p.m. of all houses

The voltage behavior, as shown in Figure 6.3, is clearly nonlinear. And, the voltage drop distance between households decreases as we go down the node. This explains that the voltage trend is associated with the line impedance as well as the feeder loading, rather than a particular source voltage. This behavior is explain by:

$$\Delta\bar{\mathbf{V}} \approx \frac{RP + XQ}{\bar{\mathbf{V}}}$$

See the derivation of the above equation in the previous Section Equation 4.14.

6.0.3 Thinking out loud

The question that we try to answer in this section is:

- Can an inverter at say, house 15, raise its voltage from (e.g., 0.92) up toward a target value (1.0 p.u.)?

Not really. Figure 6.4 shows the behavior of a five kVA inverter installed at house 15. The inverter kept providing real power the whole day, resulting in $V_{max} = 0.98$. The constant injection of Q is due to the fact that the target voltage, $V_{target} = 1.00\ p.u.$ was never reached. Note that there is a slight change inthe head of the feeder (Node 4 at 2400 V). The change is almost zero. Also, the residential head (the secondary side of the transformer at 120 V) sees a low voltage change as well, at 0.006 V.

6.0.3.1 Determining the relationship between Q and the Voltage

We already know the relationship between the voltage and reactive power. From the Jacobian submatrix,

$$\bar{J}_4 = \frac{\partial \bar{Q}}{\partial |V|}$$

Sweeping the reactive power (Q) from $-5\ kVAr$ to $+5\ kVAr$ on a 5 kVA inverter at house 15 causes the voltage at that house to dropt first, because we are absorbing Q ($-Q$). Note that at the top plot of Figure 6.5 the house voltage goes from 0.88 p.u. to 0.92. However, doing the unreasonable thing and have that inverter inject reactive power (Q) all day long would improve the house15 voltage to 0.98 p.u., as shown in Figure 6.4.

The mid plot in Figure 6.5 shows the voltage at the distribution transformer in blue and the substation transformer in orange. The y-axis the voltage difference between the basecase and the Q sweep.

Note that the voltage deviation is worest at 20:46, so the Figure 6.5 is only running the simulation at that time.

The lower plot, showing the voltages at each house, as we sweep Q at the inverter at house 15. Note that the voltage difference between, for example, $+Q = 5\ kVAr$ and $Q = 0\ kVAr$ is larger at house 15 than at the first house. In fact, this difference decreases as we get closer to the head of the branch. Again, referring the Jacobian matrix, particularly, J_4, which describes the voltage/Q relationship:

$$\frac{\partial \bar{Q}_{15}}{\partial |V_i|} \quad \text{where, } i \ \in \{1, 2, 3, \cdots, 15\}$$

We can check the results in Figure 6.5 against our equations, as derived in Equation 4.14

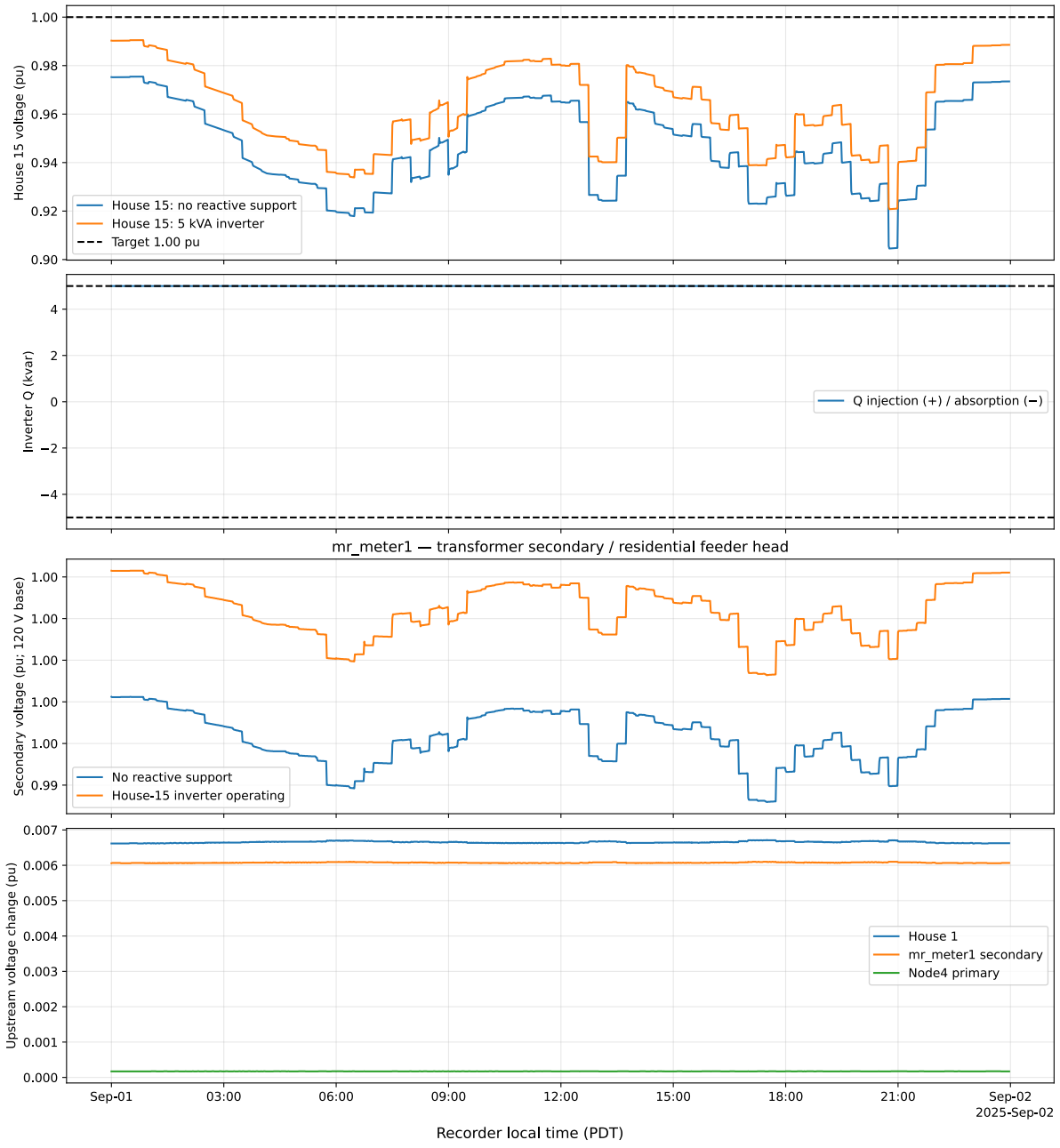


Figure 6.4: Inverter Operation at House 15

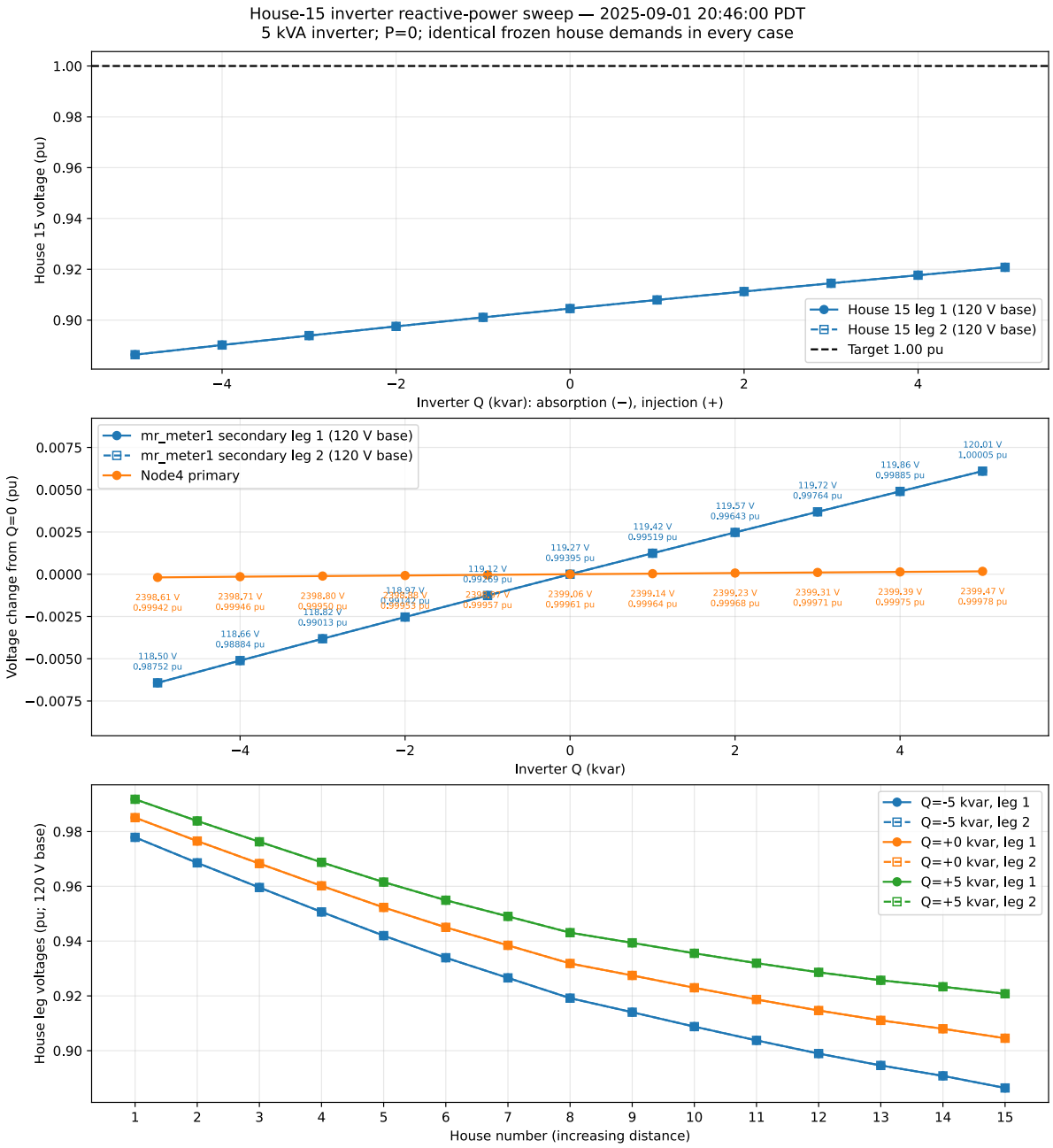


Figure 6.5: Inverter Operation at House 15

Note: GridLAB-D uses another version of the Jacobian Matrix, that is the Three-phase Current Injection Method (TCIM). The next few chapters explain the TCIM works.

6.0.3.2 Verification

6.0.3.2.1 Transformer Voltage:

In the GridLAB-D model, the distribution transformer is rated at 50 kVA, resistance at 1%, and reactance is at 6%. Using Equation 4.14:

- $r = 0.01\ p.u.$
- $x = 0.06\ p.u.$
- $S = 50\ kVA$
- $Q \in \{-5, 0, 5\}\ kVAr$
- $V = 240\ V$

Therefore:

Calculating the base z value:

$$Z_{base} = \frac{V_{base}^2}{S_{base}} = \frac{(240)^2\ V}{50\ KVA} = 1.15\ \Omega$$

Then the actual **x** becomes:

$$x_{actual} = 0.06\ p.u. \times 1.15\ \Omega = 0.069\ \Omega$$

Calculating the actual r value:

$$r_{actual} = 0.01\ p.u. \times 1.15\ \Omega = 0.0115\ \Omega$$

Since only reactive power is being injected, we can neglect P **and** r Plugging in values in the ΔV equation:

$$\Delta V = \frac{RP + XQ}{V} \approx \frac{XQ}{V}$$
$$\Delta V = \frac{(0.069\ \Omega)(5\ kVAr)}{240\ V} = 1.44\ V$$

Converting the voltage back to per unit:

$$V_{p.u.} = \frac{V_{act}}{V_{base}} = \frac{1.44\ V}{240\ V} \approx 0.006 \approx 0.6\%$$

Note that the 50 kVA is at the transformer, not house 15.

6.0.3.2.2 Understanding the Jacobian used in this experiment

The Jacobian describes how small changes in the assumed bus voltages change the calculated electrical quantities. To interpret the matrix printed by GridLAB-D, we first need to identify those voltage values and the quantities the solver is trying to balance.

6.0.3.2.2.1 Step 1: Describe one voltage using two numbers

Chapter 2 uses a complex voltage, written as $\bar{V}$. The house-15 leg-1 voltage in the zero-Q case at 20:46 is

$$\bar{V}_{15} = 108.363349 - j6.272459 \text{ V}.$$

This voltage has a real part, 108.363349 V, and an imaginary part, −6.272459 V. The subscripts R and I identify these two parts:

$$V_R = 108.363349 \text{ V}, \qquad V_I = -6.272459 \text{ V},$$

so the same voltage can be written as

$$\bar{V} = V_R + jV_I.$$

This is the rectangular representation of the voltage. The magnitude and angle provide another representation of exactly the same voltage:

$$|\bar{V}_{15}| = 108.544733 \text{ V}, \qquad \delta_{15} = -3.312789^\circ.$$

GridLAB-D's solver in this experiment works with the real and imaginary parts. A conventional angle/magnitude power-flow calculation works with the magnitude and angle instead. The physical voltage is the same; the two calculations use different pairs of numbers to describe it.

The notation $e + jf$ means $V_R + jV_I$: e, which is the real part and f is the imaginary part. We will use V_R and V_I here to make their meanings visible.

The notation $\text{Im}\{\bar{V}\}$ means **take the imaginary part of the voltage**. Thus:

$$\text{Im}\{\bar{V}_{15}\} = -6.272459 \text{ V}.$$

6.0.3.2.2.2 Step 2: Calculate the specified current and the current mismatch

The specified power at a connection is generation minus load consumption:

$$\bar{S}_{\text{specified}} = (P_{\text{inverter}} - P_{\text{house}}) + j(Q_{\text{inverter}} - Q_{\text{house}}).$$

Using the current iteration's voltage estimate, chapter 2's complex-power equation gives the specified device current:

$$\bar{I}_{\text{specified}} = \frac{\bar{S}^*_{\text{specified}}}{\bar{V}^*} = \frac{(P_{\text{inverter}} - P_{\text{house}}) - j(Q_{\text{inverter}} - Q_{\text{house}})}{\bar{V}^*}.$$

This is the net current injected by the devices at the connection. Generation is positive; consumption is negative. For constant-power devices, the power is specified, but the current is recalculated using the voltage estimate at each iteration.

For example, a **single 120 V connection** with a 1,200 W load, no reactive demand, and no inverter generation has

$$\bar{I}_{\text{specified}} = \frac{-1200}{120} = -10 \text{ A}.$$

The negative value means that the house consumes current rather than injecting it. This example uses one 120 V connection; the actual house model has two legs, so 240 V.

The network also determines a current from the assumed bus voltages. At house 15, the current leaving through the service line is

$$\bar{I}_{\text{network,15}} = \frac{\bar{V}_{15} - \bar{V}_{\text{tap15}}}{\bar{Z}_{\text{service15}}}.$$

For a connection with several lines, the network current is the sum of the currents leaving through those lines.

The **current mismatch (instead of power mismatch)** is specified device injection minus calculated network current:

$$\boxed{\Delta\bar{I} = \bar{I}_{\text{specified}} - \bar{I}_{\text{network}}.}$$

Continuing the 1,200 W example, if the assumed voltages give $\bar{I}_{\text{network}} = -9$ A, then

$$\Delta\bar{I} = (-10) - (-9) = -1 \text{ A}.$$

If the network current is -10 A, the mismatch is zero and current balance is satisfied. Collecting the real and imaginary mismatches at all connections produces the **current mismatch vector**.

The ordinary power-flow mismatch vector compares specified and calculated powers:

$$\begin{bmatrix} \Delta P_i \\ \Delta Q_i \end{bmatrix} = \begin{bmatrix} P_{i,\text{specified}} - P_{i,\text{calculated}} \\ Q_{i,\text{specified}} - Q_{i,\text{calculated}} \end{bmatrix}.$$

The current-injection formulation instead compares specified and network currents. Written with the imaginary components first, matching our GridLAB-D block arrangement, this is

$$\begin{bmatrix} \Delta I_{i,Im} \\ \Delta I_{i,R} \end{bmatrix} = \begin{bmatrix} I_{i,Im,\text{specified}} - I_{i,Im,\text{network}} \\ I_{i,R,\text{specified}} - I_{i,R,\text{network}} \end{bmatrix}.$$

Both parts should be approximately zero when the currents balance. The calculation below shows this for house 15.

Worked calculation: house 15 at 20:46, with inverter Q = 0

We start with the solved zero-Q case, before changing the inverter output. Every voltage and current below refers to **one balanced 120 V leg equivalent**. We do not mix whole-house power with a single-leg voltage.

1. Read the house power. In the zero-Q simulation file, house 15 has

```
constant_power_12 6623.66668249131+0j;
```

This is the whole-house load across the 240 V connection:

$$P_{\text{house}} = 6623.666682 \text{ W}, \qquad Q_{\text{house}} = 0 \text{ var}.$$

For the balanced single-leg calculation, divide the house power equally between the two legs:

$$P_{\text{house,leg}} = \frac{6623.666682}{2} = 3311.833341 \text{ W}, \qquad Q_{\text{house,leg}} = 0 \text{ var}.$$

Initially the inverter has $P_{\text{inv}} = 0$ W and $Q_{\text{inv}} = 0$ var, so its contribution to either leg is zero.

2. Read the house voltage and take its conjugate. The zero-Q voltage at house 15 is

$$\bar{V}_{15} = 108.363349 - j6.272459 \text{ V}.$$

The specified-current equation uses the **conjugate** of this voltage. Taking the conjugate changes the sign of the imaginary part:

$$\bar{V}_{15}^{*} = 108.363349 + j6.272459 \text{ V}.$$

3. Calculate the specified current. Substituting the single-leg power and voltage gives

$$\bar{I}_{\text{specified},15} = \frac{(0 - 3311.833341) - j(0 - 0)}{108.363349 + j6.272459} = \frac{-3311.833341}{108.363349 + j6.272459}.$$

Dividing the complex numbers gives

$$\boxed{\bar{I}_{\text{specified},15} \approx -30.460244 + j1.763148 \text{ A}.}$$

The denominator is voltage, not line impedance. This current comes from the specified device power and the house-voltage estimate.

4. Read the neighboring tap voltage and service-line impedance. House 15 connects to tap 15 through the 30 ft service line:

```
tap 15 ---- service_15 ---- house 15
```

The voltage at tap 15 is

$$\boxed{\bar{V}_{\text{tap15}} = 108.463828 - j6.246562 \text{ V}.}$$

This is a different connection from house 15. Their voltages are close because the service line is short: their magnitudes are approximately 108.644 V and 108.545 V, respectively.

For the same balanced single-leg equivalent, the service impedance is

$$\boxed{\bar{Z}_{15} = 0.0032386364 + j0.0010376674 \ \Omega.}$$

This is the impedance of `service_15` only, not the whole path from the transformer. It is the balanced-leg equivalent obtained from the model's exported impedance matrix, as used in the chapter-2 verification script.

5. Calculate the network current. Current leaving house 15 toward tap 15 is

$$\bar{I}_{\text{network},15} = \frac{\bar{V}_{15} - \bar{V}_{\text{tap15}}}{\bar{Z}_{15}}.$$

These voltages are used directly, **without taking their conjugates**. First subtract their real parts:

$$108.363349 - 108.463828 = -0.100479 \text{ V}.$$

Then subtract their imaginary parts:

$$-6.272459 - (-6.246562) = -0.025897 \text{ V}.$$

Therefore,

$$\bar{V}_{15} - \bar{V}_{\text{tap15}} = -0.100479 - j0.025897 \text{ V}.$$

Dividing this voltage difference by the service impedance gives

$$\bar{I}_{\text{network,15}} = \frac{-0.100479 - j0.025897}{0.0032386364 + j0.0010376674} \approx -30.460147 + j1.763243 \text{ A}.$$

This closely matches the specified current. Both currents use the same convention: current injected into the network, or leaving the house through the service line, is positive.

6. Calculate the current mismatch. Subtract network current from specified current:

$$\begin{aligned}\Delta\bar{I}_{15} &= \bar{I}_{\text{specified,15}} - \bar{I}_{\text{network,15}} \\ &\approx (-30.460244 + j1.763148) - (-30.460147 + j1.763243) \\ &\approx -0.000097 - j0.000095 \text{ A} \approx 0.\end{aligned}$$

The small remaining difference reflects the precision of the printed voltages and impedances. These inputs came from a solved operating point, so the specified current and network current should agree approximately. Rounded to two decimal places, both are $-30.46 + j1.76$ A.

The house powers, branch voltages and impedances for this calculation are recorded in `gld/verify_ch2/branch_measurements.csv`, selecting `q_injected_kvar = 0` and `branch = service_15` for the service-line values. The house demand is specified in `gld/results/house15_q_sweep/q_+00000var/case.glm`.

6.0.3.2.2.3 Step 3: Use the current mismatch to correct the voltages

The Jacobian relates voltage corrections to the current mismatch. The Newton–Raphson calculation is

$$\boxed{\mathbf{J}^{(k)}\Delta\mathbf{V}^{(k)} = \Delta\mathbf{I}^{(k)}.}$$

Here, k identifies the current iteration. The vector $\Delta\mathbf{I}^{(k)}$ contains the current mismatches just calculated. The vector $\Delta\mathbf{V}^{(k)}$ contains the real and imaginary voltage corrections. After solving for those corrections, the voltages are updated:

$$\mathbf{V}^{(k+1)} = \mathbf{V}^{(k)} + \Delta\mathbf{V}^{(k)}.$$

GridLAB-D then recalculates the currents and mismatches using the updated voltages. This repeats until the solver's convergence checks are satisfied.

The Jacobian accounts for how both network current and specified device current change with voltage. Its entries are derivatives of network current minus specified device current, with respect to the voltage components. The entries have units A/V. This includes the voltage dependence of constant-power device current.

All connected voltage corrections are calculated together. For example, changing house-15 voltage changes service-line current at both house 15 and tap 15.

6.0.3.2.2.4 Step 4: Identify what TCIM means

TCIM is the three-phase current-injection method. It is the name of GridLAB-D's current-balance power-flow formulation. It can also handle the single-phase and split-phase connections in this model; the name does not mean that every connection in this experiment has three phases.

The settings `solver_method NR` and `NR_solver_algorithm TCIM` describe two related choices:

- **TCIM** identifies the equations and voltage representation: current balance using real/imaginary voltage components.
- **NR** identifies how those equations are solved: repeated Newton–Raphson voltage corrections.

Consequently, the printed matrix describes changes in **current mismatch** caused by changes in **real/imaginary voltage**, rather than changes in P and Q caused by changes in voltage angle and magnitude.

6.0.3.2.2.5 Step 5: Write the full Jacobian with each quantity defined

First, consider the conventional power-flow Jacobian used when discussing $\mathbf{J}_4$. The quantities involved are:

Symbol	Meaning
$\mathbf{P}$	List of calculated real-power injections at the included connections
$\mathbf{Q}$	List of calculated reactive-power injections at those connections
δ	List of voltage angles
$\mathbf{v}$	List of voltage magnitudes, $v_i = \lvert\bar{V}_i\rvert$

Power injection is positive for generation and negative for consumption. Bold symbols collect values from multiple connections into vectors. Each Jacobian entry is a partial derivative: one quantity changes while the other independent voltage variables remain fixed.

The four blocks are

$$\boxed{\mathbf{J}_{PQ} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{J}_2 \\ \mathbf{J}_3 & \mathbf{J}_4 \end{bmatrix} = \begin{bmatrix} \dfrac{\partial \mathbf{P}}{\partial \delta} & \dfrac{\partial \mathbf{P}}{\partial \mathbf{v}} \\ \dfrac{\partial \mathbf{Q}}{\partial \delta} & \dfrac{\partial \mathbf{Q}}{\partial \mathbf{v}} \end{bmatrix}.} \tag{6.1}$$

For a Newton calculation, $\Delta\mathbf{P}$ and $\Delta\mathbf{Q}$ are the specified power injections minus the injections calculated from the current voltage estimate. The required voltage corrections satisfy

$$\begin{bmatrix} \mathbf{J}_1 & \mathbf{J}_2 \\ \mathbf{J}_3 & \mathbf{J}_4 \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta\mathbf{v} \end{bmatrix} = \begin{bmatrix} \Delta\mathbf{P} \\ \Delta\mathbf{Q} \end{bmatrix}.$$

In particular, $(\mathbf{J}_4)_{15,15}$ describes how calculated Q_{15} changes with house-15 voltage magnitude **while all angles and other magnitudes remain fixed**. The inverter sweep does not hold those other quantities fixed. Its measured voltage response therefore cannot be obtained by simply taking the reciprocal of that one entry. The other blocks also participate in the voltage response.

For GridLAB-D, the voltage variables are the real and imaginary components, rather than magnitude and angle. The mismatch vector contains imaginary-current mismatches first and real-current mismatches second when grouped into the following block form:

$$\boxed{\begin{bmatrix}\Delta\mathbf{I}_{Im}\\ \Delta\mathbf{I}_{R}\end{bmatrix} = \underbrace{\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{bmatrix}}_{\mathbf{J}_{\mathrm{TCIM}}} \begin{bmatrix}\Delta\mathbf{V}_{R}\\ \Delta\mathbf{V}_{Im}\end{bmatrix}.} \tag{6.2}$$

The subscript R identifies real parts, and Im identifies imaginary parts. Each block describes a particular current/voltage relationship:

$$\mathbf{A} = \frac{\partial(\mathbf{I}_{\mathrm{network},Im} - \mathbf{I}_{\mathrm{specified},Im})}{\partial \mathbf{V}_{R}}, \quad \mathbf{B} = \frac{\partial(\mathbf{I}_{\mathrm{network},Im} - \mathbf{I}_{\mathrm{specified},Im})}{\partial \mathbf{V}_{Im}},$$
$$\mathbf{C} = \frac{\partial(\mathbf{I}_{\mathrm{network},R} - \mathbf{I}_{\mathrm{specified},R})}{\partial \mathbf{V}_{R}}, \quad \mathbf{D} = \frac{\partial(\mathbf{I}_{\mathrm{network},R} - \mathbf{I}_{\mathrm{specified},R})}{\partial \mathbf{V}_{Im}}.$$

The specified-minus-network current mismatch is on the left of Equation 6.2. The Jacobian describes the voltage corrections needed to close that difference. The blocks describe current/voltage derivatives; $\mathbf{D}$ is not the conventional $\mathbf{J}_4 = \partial\mathbf{Q}/\partial\mathbf{v}$.

For a three-bus system with one fixed slack bus and two single-phase PQ buses, the four voltage corrections belong to buses 2 and 3:

$$\begin{bmatrix}\Delta I_{2,Im}\\ \Delta I_{3,Im}\\ \Delta I_{2,R}\\ \Delta I_{3,R}\end{bmatrix} = \begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{bmatrix} \begin{bmatrix}\Delta V_{2,R}\\ \Delta V_{3,R}\\ \Delta V_{2,Im}\\ \Delta V_{3,Im}\end{bmatrix}.$$

Each block is 2×2, so this non-slack system is 4×4. The current mismatches are calculated from the current voltage estimate, and the voltage corrections are added to obtain the next estimate.

In this three-bus example, the Jacobian matrix becomes:

$$\underbrace{\left[\begin{array}{cc|cc} \frac{\partial I_{2,Im}}{\partial V_{2,R}} & \frac{\partial I_{2,Im}}{\partial V_{3,R}} & \frac{\partial I_{2,Im}}{\partial V_{2,Im}} & \frac{\partial I_{2,Im}}{\partial V_{3,Im}} \\ \frac{\partial I_{3,Im}}{\partial V_{2,R}} & \frac{\partial I_{3,Im}}{\partial V_{3,R}} & \frac{\partial I_{3,Im}}{\partial V_{2,Im}} & \frac{\partial I_{3,Im}}{\partial V_{3,Im}} \\ \hline \frac{\partial I_{2,R}}{\partial V_{2,R}} & \frac{\partial I_{2,R}}{\partial V_{3,R}} & \frac{\partial I_{2,R}}{\partial V_{2,Im}} & \frac{\partial I_{2,R}}{\partial V_{3,Im}} \\ \frac{\partial I_{3,R}}{\partial V_{2,R}} & \frac{\partial I_{3,R}}{\partial V_{3,R}} & \frac{\partial I_{3,R}}{\partial V_{2,Im}} & \frac{\partial I_{3,R}}{\partial V_{3,Im}} \end{array}\right]}_{J = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]}$$

6.0.3.2.2.6 Step 6: Count the numbers stored in this model

Each complex voltage requires a real and an imaginary part. Each house has two voltage legs, so it contributes four real numbers:

$$V_{R,1}, \quad V_{R,2}, \quad V_{I,1}, \quad V_{I,2}.$$

The exported bus mapping gives the complete count:

Connections	Number	Real numbers per connection	Total
Fixed upstream source, phase A	1	2	2
Node4 primary, phase A	1	2	2
Transformer-secondary meter, two legs	1	4	4
House taps, two legs each	15	4	60
Houses, two legs each	15	4	60
Total			**128**

The source voltage is fixed. GridLAB-D nevertheless retains its two positions in the exported matrix. It uses large diagonal entries (10^{10} in this dump) and skips source-voltage updates during the normal solve. Those rows are special source constraints, rather than ordinary unknown-bus current-balance equations. Excluding the source's two fixed voltage components would leave 126 values to solve for.

The matrix is square because it supplies one equation for each stored voltage component. If all real voltage components were placed first and all imaginary components second, its four blocks would each be 64×64.

The raw CSV instead stores components together by bus. At a split-phase bus, its columns are ordered as

$$[V_{R,1},\ V_{R,2},\ V_{I,1},\ V_{I,2}],$$

and its current-mismatch rows are ordered as

$$[\Delta I_{1,Im},\ \Delta I_{2,Im},\ \Delta I_{1,R},\ \Delta I_{2,R}].$$

Consequently, the four global derivative blocks cannot be extracted simply by cutting the raw CSV into four equal rectangles. The rows and columns first need to be grouped according to their meanings.

6.0.3.2.2.7 Step 7: Read the house-15 numerical block

At 20:46 PDT with inverter $Q = 0$, the exported matrix assigns indices 90–93 to house 15. The four columns correspond to its two real voltage parts followed by its two imaginary voltage parts. The four rows correspond to its two imaginary current-balance equations followed by its two real current-balance equations.

The house-15 rows and house-15 columns contain

$$\mathbf{J}_{\mathrm{TCIM}}^{(15,15)} = \left[\begin{array}{rrrr} -71.195871 & -18.508666 & 191.378894 & 88.926892 \\ 18.508666 & 71.195871 & -88.926892 & -191.378894 \\ \hline 191.098270 & 88.926892 & 71.212115 & 18.508666 \\ -88.926892 & -191.098270 & -18.508666 & -71.212115 \end{array}\right]. \tag{6.3}$$

For example, the first row and third column contain 191.378894 A/V. This entry describes the change in leg-1 imaginary network current minus specified device current caused by a small change in the leg-1 imaginary voltage, with the other voltage components fixed. A 0.001 V increase would therefore increase that calculated current difference by approximately 0.191379 A before other voltages are adjusted.

This 4×4 block describes house 15's own voltage components. The complete house-15 equations also contain entries in the columns for tap 15, because the service-line current depends on both ends of that line. Those neighboring entries are one reason the complete network matrix is needed.

7 Quantify the decoupling range.

Develop a sensitivity-based model that determines the decoupling range as a function of inverter count, location, phase assignment and rating.

7.1 Setup

1. Calibrate the base case. At a Node 4 voltage of 1.00 pu and the worst hour (20:46), the last house should sit between 0.95 and 0.97 pu. Adjust loading, secondary length or the conductor value until it does.
2. Fix two operating points with frozen loads, as you did for Figure 6.5:
    - **Heavy load (20:46):** sets the lower edge of the range, where houses hit 114 V.
    - **Light load (the daily minimum):** sets the upper edge, where houses hit 126 V.
3. Define "pass": every house, both legs, between 0.95 and 1.05 pu.

Baseline: no inverter

4. Sweep the swing voltage from about 0.95 to 1.05 pu in small steps (0.005 pu).
5. Record two numbers:
    - the lowest primary voltage at which every house still passes at heavy load;
    - the highest primary voltage at which every house still passes at light load.

The band between them is the baseline range. It may be very narrow, or even empty.

One inverter

6. Add one 5 kVA inverter at house 15.
    - **Heavy load:** inject +2.2 kvar, the IEEE 1547 limit of 44%.
    - **Light load:** absorb −2.2 kvar.
    - Also run ± 5 kvar as an upper bound.
7. Repeat the sweep and record the new lowest and highest primary voltages.

8. Decoupling range = new range − baseline range. Need to write it as one sentence: “With one 5 kVA inverter at house 15, the lower edge moves from X to Y pu.”
9. Note which house is now the limit. Once house 15 is fixed, another house (for example 14 or 13) may become the one that fails first. This shows why one inverter is rarely enough.

Extend it

10. Location: move the single inverter from house 1 to house 15 and record the range at each position.
11. Count: add inverters one at a time, farthest house first, from 1 to 15, and record the range each time.
12. Rating: repeat step 7 with 3, 5, 7.6 and 10 kVA inverters.

Deliverables

- A table with one row per case (baseline, location, count, rating), giving the lower edge, upper edge and limiting house.
- Three plots: range against location, range against count, and range against rating.

7.2 Results

Calibrate the base case. At a Node 4 voltage of 1.00 pu and the worst hour (20:46), the last house should sit between 0.95 and 0.97 pu. Adjust loading, secondary length or the conductor value until it does.

main_segment_ft	node4_pu	house	v_at_2046_pu	day_min_pu	day_min_time	in_target
100	1	15	0.90497	0.90497	2025-09-01 20:46:00 PDT	False
40	1	15	0.96061	0.96061	2025-09-01 20:46:00 PDT	True

Fix two operating points with frozen loads, as you did for Figure 6.5: Heavy load (20:46): sets the lower edge of the range, where houses hit 114 V. Light load (the daily minimum): sets the upper edge, where houses hit 126 V.

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point	timestamp	node4_pu	house	voltage_1_pu	voltage_2_pu
heavy	2025-09-01 20:46:00	1	1	0.991533	0.991533
heavy	2025-09-01 20:46:00	1	2	0.988283	0.988283
heavy	2025-09-01 20:46:00	1	3	0.985175	0.985175
heavy	2025-09-01 20:46:00	1	4	0.982058	0.982058
heavy	2025-09-01 20:46:00	1	5	0.979025	0.979025
heavy	2025-09-01 20:46:00	1	6	0.976242	0.976242
heavy	2025-09-01 20:46:00	1	7	0.9738	0.9738
heavy	2025-09-01 20:46:00	1	8	0.971092	0.971092
heavy	2025-09-01 20:46:00	1	9	0.969667	0.969667
heavy	2025-09-01 20:46:00	1	10	0.967975	0.967975

7.2.1 Baseline: no inverter

1. Sweep the swing voltage from about 0.95 to 1.05 pu in small steps (0.005 pu). Logging two numbers:

    - the lowest primary voltage at which every house still passes at heavy load;
    - the highest primary voltage at which every house still passes at light load.

Figure 7.1 shows the per unit house voltages on the y-axis and the per unit swing node voltages sweep. The sweep is intentional such that we change the voltages at the swing node from the 0.95 to 1.05 at 0.005 p.u. voltage step.

As expected, house 15 (the furthest from the head of the line) has the lowest voltage drop. The blue line, as well as the shaded area, shows a heady load case, wherein the load profile is multiplied by 1.25% of the original ResStock dataset. The orange line, on the other hand, shows a light load, wherein the demand profile is kept the same. The big rectangular shaded area is the $\pm$ 5% region. This range is here for testing.

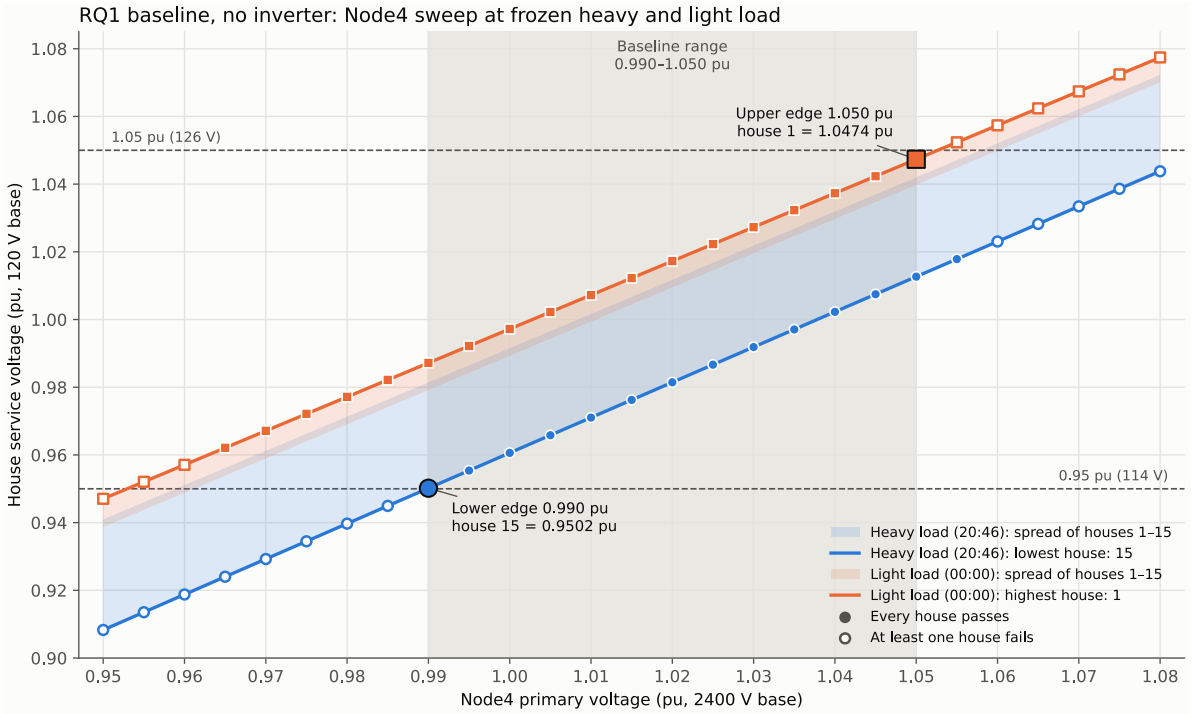


Figure 7.1: Baseline Sweep with no inverters

8 Introduction

References

Gan, Lingwen, and Steven H Low. 2014. "Convex Relaxations and Linear Approximation for Optimal Power Flow in Multiphase Radial Networks." *2014 Power Systems Computation Conference*, 1–9.